

LAG-1 Session 8 v0.1 outline — Explicit 32×32 γ -matrix construction in Hua coordinates

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LAG-1 Session 8 v0.1 — Explicit 32×32 γ -matrix construction in Hua coordinates of D_{IV}^5

1. Goal

Construct the 32×32 complex γ -matrices $\gamma^{\{z_i\}}$ ($i = 1, \dots, n_C = 5$) and $\gamma^{\{\bar{z}_j\}}$ ($j = 1, \dots, n_C = 5$) explicitly in Hua coordinates on D_{IV}^5 , satisfying the Clifford anti-commutation relations (T2349):

$$\begin{aligned}\{\gamma^{\{z_i\}}, \gamma^{\{\bar{z}_j\}}\} &= 2 g^{\{i\bar{j}\}} \quad (\text{Bergman metric}) \\ \{\gamma^{\{z_i\}}, \gamma^{\{z_j\}}\} &= 0 \\ \{\gamma^{\{\bar{z}_i\}}, \gamma^{\{\bar{z}_j\}}\} &= 0\end{aligned}$$

The matrices act on the 32-dim Dolbeault-spinor bundle $S = \Lambda^* T^{\{0,1\}} D_{IV}^5$, with chirality split $S = S^+ \oplus S^-$ (dim 16 each).

This converts T2349 from D-tier algebraic (anti-commutation relations satisfied abstractly) to D-tier explicit (closed-form matrices in Hua coordinates).

2. Sub-tasks (scope: ~1-2 weeks focused)

2.1 Hua coordinates on D_{IV}^5

- D_{IV}^5 in standard Hua coordinates: $z \in \mathbb{C}^5$ with constraint $|z|^2 < 1 + |z^{\bar{z}}z|^2$ (type-IV domain)
- Bergman metric $g_{\{i\bar{j}\}} = \partial_i \bar{\partial}_{\bar{j}} \log K_B$ with $K_B = c \cdot D(z, \bar{z})^{\{-7/2\}}$ (T2334)
- Inverse metric $g^{\{i\bar{j}\}}$ required for Clifford anti-commutation

2.2 Dolbeault spinor bundle basis

- 32 basis sections of $\Lambda^* T^{\{0,1\}} D_{IV}^5$: from $|\Omega\rangle$ (no anti-holomorphic forms) through 5 single forms through 10 double forms through 10 triple forms through 5 quadruple forms through 1 maximal form $|d\bar{z}^1 \wedge d\bar{z}^2 \wedge d\bar{z}^3 \wedge d\bar{z}^4 \wedge d\bar{z}^5\rangle$
- Total: $2^{n_C} = 32 = \text{rank}^{n_C}$ basis sections
- Chirality split: even-degree (16 sections) vs odd-degree (16 sections)

2.3 γ -matrix entries

- $\gamma^{\{z_i\}}$: holomorphic generator acts as $\varepsilon(d\bar{z}^i)/\sqrt{2}$ — wedge product with $d\bar{z}^i$ — adds one anti-holomorphic form. Matrix maps (k-form sector) $\rightarrow$ (k+1)-form sector.
- $\gamma^{\{\bar{z}_j\}}$: anti-holomorphic generator acts as $\sqrt{2} \cdot \iota(\partial/\partial z^j)$ — interior product with $\partial/\partial z^j$ — removes one anti-holomorphic form. Matrix maps (k+1)-form sector $\rightarrow$ (k-form sector).
- Construction is direct: each γ -matrix has exactly 16 non-zero entries (one per spinor basis state where the wedge/contraction is defined).

2.4 Verification: anti-commutation relations

- Direct matrix computation: $\gamma^{\{z_i\}}\gamma^{\{\bar{z}_j\}} + \gamma^{\{\bar{z}_j\}}\gamma^{\{z_i\}} = 2g^{\{ij\}} \cdot I_{\{32\}}$
- $\gamma^{\{z_i\}}\gamma^{\{z_j\}} + \gamma^{\{z_j\}}\gamma^{\{z_i\}} = 0$
- $\gamma^{\{\bar{z}_i\}}\gamma^{\{\bar{z}_j\}} + \gamma^{\{\bar{z}_j\}}\gamma^{\{\bar{z}_i\}} = 0$

2.5 Verification: chirality split

- Define $\Gamma_5 = \gamma^{\{z_1\}}\gamma^{\{\bar{z}_1\}}\gamma^{\{z_2\}}\gamma^{\{\bar{z}_2\}}\gamma^{\{z_3\}}\gamma^{\{\bar{z}_3\}}\gamma^{\{z_4\}}\gamma^{\{\bar{z}_4\}}\gamma^{\{z_5\}}\gamma^{\{\bar{z}_5\}}$ or appropriate normalization
- Verify $\Gamma_5^2 = 1$ (chirality matrix)
- Verify Γ_5 has eigenvalues ± 1 with eigenspaces dim 16 each

2.6 Verification: Lichnerowicz formula explicit

- Compute D^2 explicitly via $D = \gamma^{\{z_i\}}\nabla_{\{z_i\}} + \gamma^{\{\bar{z}_j\}}\nabla_{\{\bar{z}_j\}}$
- Verify $D^2 = \nabla\nabla + R/4 = \nabla\nabla - n_C \cdot g/4$ (T2352)
- Connect to Wallach K-type spectrum (T2351)

3. BST integer reading of the construction

Quantity	Value	BST primary form
Spinor dim	32	$\text{rank}^{\{n_C\}} = 2^5$
Clifford generator count	10	$\text{rank} \cdot n_C = \text{dim}_R D_{IV}^5$
Chirality multiplicity	16	$2^{\{n_C-1\}} = 2^4$
Bergman metric components	$25 = n_C^2$	n_C^2 components in metric tensor
Lichnerowicz shift $R/4$	$-35/4$	$-n_C \cdot g/4$ (BST primary product)
First-excited eigenvalue $\lambda_W(1,0)$	$C_2 = 6$	Bergman Casimir

4. Tier path

Current (Session 7 close): T2349 D-tier algebraic, T2351 D-tier Wallach spectrum, T2352 D-tier Lichnerowicz. Mechanism is operator-identified (Paper #118 v0.1) but the matrices are abstract.

After Session 8 close: T2349 promotes from “Clifford structure satisfied abstractly” to “explicit 32×32 matrices in Hua coordinates with all anti-commutation relations verified at matrix entry level.” This is the “D-tier explicit” promotion.

Downstream enabled by Session 8: - Session 9: heat kernel evaluation $\text{Tr}(e^{-(tD^2)})$ explicit (multi-week) - Session 10: index theorem / chiral anomaly explicit (multi-month) - Per-flavor K-type SM assignment (multi-month, requires Sessions 9+10) - m_p/m_e precision derivation to numerical (multi-week, Bergman volume integral)

5. Verification strategy

- Toy 30xx (LAG-1 S8 verification): claim toy number; build 32×32 numpy matrices for γ -matrices; verify all anti-commutation relations to machine precision; verify chirality matrix $\Gamma_5^2 = I$; verify D^2 Lichnerowicz formula on test sections.
- Theorem T2363 (or successor): T2349 promoted to explicit-matrix form. Cross-references to T2351, T2352.
- Paper #118 v0.2 update: Section 3 (Clifford algebra) expanded with explicit matrix construction. Paper #118 v0.2 expected closure target: end of Session 8.

6. Open beyond Session 8

The Session 8 deliverable closes the *matrix-level* construction. Beyond Session 8 the following remain (per Section 9.x): - Bergman volume normalization integral for m_p/m_e numerical precision (multi-week) - Per-flavor

K-type SM fermion assignment (multi-month) - Heat-kernel $\text{Tr}(e^{(-tD^2)})$ (multi-week, Session 9 candidate) - Index theorem in 5D / chiral anomaly (multi-month, Session 10 candidate) - Connection to LAG-2 dimensional reduction (multi-month, in parallel)

7. Time horizon

Per Section 9.x estimate: ~1-2 weeks focused work for Session 8 v1.0. v0.1 outline (this document) ~30 minutes. v0.2 detailed sub-task plan with code stubs: ~2-4 hours. v1.0 full execution: ~1-2 weeks.

8. Calibration note

Per Cal's selection-effect audit on the density rule and Keeper's Monday K-audit on LAG-2 "STRUCTURALLY CLOSED" overclaim — the framing of Session 8 stays at "explicit matrix construction with anti-commutation verification," NOT "BST mass-gap fully derived." The explicit matrices enable downstream computations; they do not themselves close m_p/m_e precision derivation or per-flavor assignment.

Honest scoping per K43+K44+Cal Coincidence_Filter_Risk + External_Survivability_Checklist: - Closed by Session 8 v1.0: explicit Clifford structure on D_{IV}^5 in Hua coordinates, machine-verifiable. - Still open: everything downstream that uses the matrices — heat kernel, index theorem, per-flavor assignment, mass-gap precision, etc.

— Lyra, LAG-1 Session 8 v0.1 outline filed 2026-05-18 ~12:40 EDT per Keeper's pivot recommendation after Cal's density-rule walk-back.